

Advanced Machine Learning

DATA 642

Lecture 7 — Canonical Correlation Analysis (CCA)

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly, and <https://towardsdatascience.com/canonical-correlation-analysis-b1a38847219d>

Canonical Correlation Analysis (CCA)

- PCA is a dimensionality reduction technique focusing on a single data set. However, in a number of cases, one has to deal with multiple data sets which, although they may originate from different sources, are closely related.
- Canonical correlation analysis (CCA) is an old technique to process two data sets jointly. Our starting point is the fact that when two sets of random variables (two random vectors) are involved, the value of their correlation does depend on the coordinate system in which the random vectors are represented.
- The goal behind CCA is to seek a pair of linear transformations, one for each set of variables, such that after the transformation, the resulting transformed variables are maximally correlated.
- There are two central concepts to understand when doing Canonical Correlation Analysis:
 - Canonical Variables
 - Canonical Correlation

The Math

- Let us assume that we are given two sets of random variables comprising the components of two random vectors, $\mathbf{x} \in \mathbb{R}^p$ and $\mathbf{y} \in \mathbb{R}^q$ and let the corresponding sets of observations be $\mathbf{x}_n, \mathbf{y}_n$ with $n = 1, \dots, N$.
- Following a step-wise procedure, as we did for PCA, we will first compute a single pair of directions, namely $\mathbf{u}_{x,1}$ and $\mathbf{u}_{y,1}$ so that the correlation between the projections onto these directions is maximized.
- Maximizing the correlation coefficient between $\mathbf{u}_{x,1}^\top \mathbf{x}$ and $\mathbf{u}_{y,1}^\top \mathbf{y}$ with respect to the directions $\mathbf{u}_{x,1}$ and $\mathbf{u}_{y,1}$ can equivalently be cast as the following constrained optimization task

$$\begin{aligned} \max_{\mathbf{u}_x, \mathbf{u}_y} \quad & \mathbf{u}_x^\top \hat{\Sigma}_{x,y} \mathbf{u}_y \\ \text{subject to} \quad & \mathbf{u}_x^\top \hat{\Sigma}_{x,x} \mathbf{u}_x = 1, \text{ and } \mathbf{u}_y^\top \hat{\Sigma}_{y,y} \mathbf{u}_y = 1, \end{aligned} \tag{1}$$

where $\Sigma_{x,y}$ is the cross-covariance matrix.

The Math (cont.)

For CCA, two directions have to be computed and the constraints involve the weighted Σ norm instead of the Euclidean one. Moreover, in PCA the variance is maximized, while CCA cares for the correlation between the projections of the two involved vectors onto the new axes.

This is a constrained optimization problem and the corresponding Lagrangian is given by

$$\mathcal{L}(\mathbf{u}_x, \mathbf{u}_y, \lambda_x, \lambda_y) = \mathbf{u}_x^\top \hat{\Sigma}_{x,y} \mathbf{u}_y - \frac{\lambda_x}{2} (\mathbf{u}_x^\top \hat{\Sigma}_{x,x} \mathbf{u}_x - 1) - \frac{\lambda_y}{2} (\mathbf{u}_y^\top \hat{\Sigma}_{y,y} \mathbf{u}_y - 1). \quad (2)$$

Taking the gradient with respect to $\mathbf{u}_x$ and $\mathbf{u}_y$ and equating to zero we obtain $\lambda_x = \lambda_y = \lambda$ and $\hat{\Sigma}_{x,y} \mathbf{u}_y = \lambda \hat{\Sigma}_{x,x} \mathbf{u}_x$ and $\hat{\Sigma}_{y,x} \mathbf{u}_x = \lambda \hat{\Sigma}_{y,y} \mathbf{u}_y$.

Solving the latter of the two with respect to $\mathbf{u}_y$ and substituting into the first one, we finally get

$$\hat{\Sigma}_{x,y} \hat{\Sigma}_{y,y}^{-1} \hat{\Sigma}_{y,x} \mathbf{u}_x = \lambda^2 \hat{\Sigma}_{x,x} \mathbf{u}_x \quad (3)$$

and

$$\mathbf{u}_y = \frac{1}{\lambda} \hat{\Sigma}_{y,y}^{-1} \hat{\Sigma}_{y,x} \mathbf{u}_x, \quad (4)$$

assuming invertibility of $\hat{\Sigma}_{y,y}$.

Furthermore, assuming invertibility of $\hat{\Sigma}_{x,x}$, too, we end up with the following eigenvalue-eigenvector problem:

$$(\hat{\Sigma}_{x,x}^{-1} \hat{\Sigma}_{x,y} \hat{\Sigma}_{y,y}^{-1} \hat{\Sigma}_{y,x}) \mathbf{u}_x = \lambda^2 \mathbf{u}_x. \quad (5)$$

Thus, the axis $\mathbf{u}_{x,1}$ is obtained as an eigenvector of the product of matrices in the above parentheses. The corresponding optimal value of the correlation is equal to $\mathbf{u}_{x,1}^\top \hat{\Sigma}_{x,y} \mathbf{u}_{y,1} = \lambda \mathbf{u}_{x,1}^\top \hat{\Sigma}_{x,x} \mathbf{u}_{x,1} = \lambda^2$.

Hence, selecting $\mathbf{u}_{x,1}$ to be the eigenvector corresponding to the maximum eigenvalue, λ^2 , results in maximum correlation.

The eigenvectors $\mathbf{u}_{x,1}$, $\mathbf{u}_{y,1}$ are known as the normalized canonical correlation basis vectors, the eigenvalue λ^2 as the squared canonical correlation, and the projections $\mathbf{u}_{x,1}^\top \mathbf{x}$ and $\mathbf{u}_{y,1}^\top \mathbf{y}$ as the canonical variates.

- In practice we will have to analyze two important sets of indicators to understand the correlations between our two datasets:
 - Interpret the Canonical Correlation: How much is each Canonical Variable from data set 1 is correlated with its counterpart in data set 2?
 - Interpret the Canonical variates: Which original variables are represented by each of the Canonical variates?